1. Introduction

During semester 2 of the academic year, I took part in a 12-week project which followed me creating a solo made game in which as a player you were a seagull set to destroy signs promoting bad environmental growth, which in turn would stop the environment from living in harmony. The reason I chose this game loop was due to the Royal Society of Arts (RSA) brief Flourishing Places. This brief is a part of RSA's Design for Life mission, which involves trying to foster a world in which people, places and the planet flourish in harmony (The RSA, 2022). Besides this, it perfectly linked with the United Nations Sustainability Development Goal (SDG) 11, which is the idea to make cities and human settlements inclusive, safe, resilient and sustainable (United Nations, 2015).

My main reason for the character and environment choice is that the oceans are an important host of life. Furthermore, over the years there have been many oil leaks from platforms that have caused much harm to these environments and animals (Greenpeace UK, n.d.). I specifically chose a seagull for the main character as I believed a seagull would be likely to want some sort of vengeance for oil leaks, as they are most affected by oil spills making them unable to fly and causing them to get hypothermia and die once covered in crude oil (International Bird Rescue, n.d.). Furthermore, I wanted my audience to have empathy for the seagull and feel they will want to play to help the seagull, which in turn will hopefully encourage players to replay. Finally, I felt that this would build my world further and be good for story telling of the world in my game.

My chosen audience for this game were 16-24-year-olds, as they are the highest percentage range of people who play games (Baker, 2025). This reason is why I chose for my character to be a seagull as younger people are more likely to care about the environment (Ballew et al., 2019) and therefore empathise with my character. I also chose for my character to be low poly and the attacks of the seagull to be humorous. I believed with empathy for the seagull and humour of the attacks to be funny it would cause strong emotions which in a moment will make something more memorable (Tyng et al., 2017). I believed with all these factors it would influence younger audiences to want change.

2. Development Process

2.1 Production Planning

For this project I first set up a Trello board which followed project managements agile workflow. This style of work meant splitting my projects large priorities into smaller manageable sprints to help with the completion of work. In terms of my project all my sprints were 2-weeks long (Figure 1), so all my work was divided into manageable tasks which I could tick off when each goal I had completed was done. During the creation of this table I made sure to follow closely to my RSA proposals example table. This meant, I had to make sure the scope of my envisioned vertical slice of my game manageable whilst still being effective in portraying my games intended message to my audience. This meant strictly sticking to this plan and designating time, so my vertical slice was complete to the level I envisioned. I planned for after each six weeks I would set a milestone to seek feedback so that I knew my project was entertaining and to ensure my work was heading in the right direction.

2.2 Software & Technical Pipeline

During the first 2 weeks of my project, I focused on setting up the basics such as the GitHub repository and Unity project with Version control so I would be able to retrieve any data so my progress would never be lost. As someone who was more confident in the creation of assets over the coding of my game, I used Blender as it was the software I was most confident with creating assets and as someone who wanted a very stylised piece of work so it would be most effective this was extremely important. To code my game, I used Visual Studio code, the reason for this was as someone who was not confident in code it was useful in identifying the lines of code which were causing issues in my code running. This allowed me to learn why certain functions worked as well as, why a certain function may have failed in my approach.

2.3 Prototype & Development Stages

During the first two weeks of my game it involved, not only the creation of the project but also the coding of my players movement as well as, the countdown timer which was used to show players how long they have left to destroy all the targets that were remaining on the map. When creating the movement, I made sure to take advantage of Unity's new input system (Figure 2) this made it easy to code in each type of controller so players would be able to play on whatever they are most comfortable with. After the movement worked, I set up a free look camera to make it easier for players to spot and notice the destroyable targets whilst moving. However, the problem with implementing the free look camera created a notable problem in moving with the compass of the character remaining the same rather than whatever direction the player faces always being north. To solve this problem, I made sure to code into my player to change its axis to face whatever direction the camera faced.

When creating the environment this was the area, I was most confident in. As the solo artist it meant I had full control of how my game would look. This allowed me to decide on a low poly art style; I did this as it meant that it would be more effective in creating a cute character and environment which I saw as more effective in making players feel empathy and laughter whilst playing my game to make it more memorable. This meant spending the sprints of weeks 3-4 and weeks 5-6 focusing on making my environment and character as well as animating the movement for my character. In weeks 3-4 it focused on my game's environment (Figures 3-4), this meant researching English seaside towns so I could get inspiration for an environment you would most likely find a seagull in. I also used inspiration from Just Cause 3 (Avalanche Studios, 2015) with red on the targets to notify audiences that it was a target they need to destroy (Figures 5-6).

When making the seagull I researched birds in media and took inspiration from the penguin (Figure 7) in Wallace & Gromit: The Wrong Trousers (Park, 1993). However, I needed to convert this look into a seagull. After much experimentation and re-iteration my seagull was made (Figure 8). During the construction process the seagull originally sat with round beadlike eyes, however after feedback I changed the eyes to look angry and to show that my character wanted vengeance. With my seagull, I set up an animator (Figure 9) on different animations the most notably being the walking, flying and attacking peck. All these animations made the game feel more real and boosted me to want to complete this project to an even more polished condition whilst staying in scope.

The gameplay systems I implemented were both attacks which relied on the same button for the player. However, I made the character switch to a flying poop like attack and a walking peck attack. With the implementation of these attacks, it meant that I had to code in damage which made use of mesh colliders so that my attacks may hit and a health script to be placed on my destroyable objects so that they disappear once their health reached zero. To win and lose the game I made the condition that when all objects were destroyed by the player before the overall timer finished, they would win however if they did not, they would lose. This meant making sure all objects were on the layer destroyable, whilst also being linked with the timer so that the condition would show the correct screen of “You Lose, Try Again” (Figure 10) or “You Win, Play Again” (Figure 11).

During the creation of the UI of this project, I made sure that no matter what device you played on there was a way to pause the game which stopped not only the timer but the movement of your character whilst easily allowing you to play again (Figure 12). Furthermore, I made sure as I said when the win and lose screens appeared there was a button which would allow the player to replay if they lost as well as playing again if they enjoyed it. Furthermore, when people played the game I made sure that players knew how much time they had left to destroy the objects around the map by leaving the number which descended in the top left as the screen as well as, making sure to show in the timer at the top middle of the screen how long the players had left to destroy those objects. A final piece of UI I implemented which related to the gameplay was I made a stamina bar when the seagull flew it would show the player it was decreasing so they knew how long they could fly for. I decided on making my seagull only capable of flying for a moment as seagulls at the beach only ever fly for a moment before landing so this to me felt more realistic, it also proved to make my game more tactical

<u>3. Feedback and Iteration</u>

During this project’s cycle I made sure to seek feedback in a multitude of ways such as testing my game to check for any bugs and so I may use my knowledge of games to see what I would prefer whilst playing so I may make my game more playable for my audience. A problem, which occurred during testing was the seagull’s animation, due to me making another mesh of the seagull in a flying state, when it was implemented into the game it meant that the animation although it worked when walking, it did not when flying. At first, I believed it was due to the animator not working; however, I realised it was due to me having a different asset for when the seagull flew and walked. To solve this issue, I re-made the skeleton of my seagull in the walking assets mesh and reanimated the flying so the animator would transition more smoothly (Figure 13). Testing also made me realise that I should add an asset which was poop shaped that would drop when the poop attack was used in the sky so players would get a good laugh from using this attack as seagulls are notorious for pooping at the beach.

Feedback I received from my peers were mainly related to the seagulls walk. The reason for this is at first when the seagull walked it followed a swing motion in the hips, however this made the character feel like it was gliding along the floor. To act on this feedback, I researched how seagulls walk and re-animated the bones of my seagull to give a more realistic movement of the character, where instead it followed a pitter patter walk style. Another point of feedback I received through playtesting from peers was that the speed the seagull walked at was too slow and the time which was left to fly was too short and took too long to recharge making travelling around the environment difficult which in turn made the objective of the game in the time frame near impossible. With a few tweaks of lengthening the time players could fly and speeding the characters walk, I was able to fix this, so my games objective was easier and felt more natural and my game felt more natural to move.

4. Critical Reflection

At the end of this projects 12-week cycle although there were many successes during this project there were failures that could be improved. For instance, a major success was that at the end of the cycle my game was playable and had both a win and loss condition which portrayed my goal of balancing humour with environmental messaging. This was proven to me at Code & Canvas through many people who tested my game enjoyed the ways in which my seagull moved and attacked. A further success I uncovered was that people were drawn to my seagulls looks and movement which allowed me to confirm that players felt empathy for my seagull. Another success I had was my character was very loved by people who played my game which meant that players empathy and support for them really grew whilst being memorable with the stupidity of the attacks and billboards being attacked.

A major failure I had whilst in production was, I ran into issues scoping issues most notably was during the development of the seagull's attacks and hit box colliders. Due to me realising to late that the meshes would not sit properly on a grid it made it difficult to position exact tiles of my game around the map. This caused problems with my mesh colliders in which if there was a flat area of the environment the seagull would encounter sudden step ups in the environment although the environment at all angles could be perceived as flat. As this issue was spotted to late it meant I had to use box carefully placed in the environment, which in turn worked but was not in a professional way I would have liked. Due to the fact of my late notice in this area it meant as I neared the end of my games production, I had less time to spend on certain areas as I had spent to much time fixing this issue. Although, this problem occurred I was able to slowly bring myself back on track as I said before, I am confident with my blender skills which allowed me to make up time. This proved successful as people grew to love my character and environments design.

5. Evaluation of Social Impact

I believe that my game was successful in portraying the RSA's goal of Flourishing places as you could clearly see through the billboards designs that as a player you would destroy. This linked with, players enjoyment of my character linked with my idea of creating empathy for players to remember my game and its message it was sending. Furthermore, it really fits with SDG 11, as the villain on the billboards "Grimey Waters" is clearly stating his love for building in ways which don't allow cities to be inclusive whilst degrading the environment, which does not allow life to live in harmony as RSA wishes. My game did prove to provide discussions about the problem of oil leakages and people did prove to show empathy and concern for the environment, specifically the creatures which are affected by such buildings or environments that do not allow them to thrive with humans in harmony as well as degrading the environment. Specifically with most people who played my game being in my target audience age range it allowed my idea of them having the idea to step up and vote for change and future projects like the Chatham Mussel Project (The RSA, 2024) which allows the world to live in harmony whilst not letting others get left behind.

6. Future Improvements

If I were to take my game and make it more polished there are a few small additions, I would add. The first would be implementing a reticle as something I noticed was that players found it difficult to aim the poop attack at targets in the air, to do this I would set up a raycast which appears only when flying. Other improvements can be seen on my Trello board I wished to implement sound effects as I know it would make my game feel more polished and complete. Furthermore, if I implemented NPC's which were under "Grimey Waters" command it would make the game more stealth like and challenging. My final minor addition which I would add is I would improve the environment and make it fuller so there were not as many blank open areas like a more realistic city.

If I were to make my game to completion, I would add more levels so that there is more playability and it encourages players to play for a longer amount of time. I would further add particle systems for destruction and rather than just having a target disappear I would make the destroyed object look different but not despawn it to make it feel more real like you as a player has destroyed something. I would also want to further increase the humour of the game and add a sprite of a poop splatter for if the poop attack is used so it will add some slapstick humour. I would further like to make the game cooperative and create another bird most likely a pigeon as they are common birds in towns as well and it will give a distinction of players.

7. Conclusion

When it comes to this project, I made sure that I was able to influence my target audience into thinking about the RSA's Flourishing Places idea as well as SDG 11. Not only this but I was able to achieve an actual vertical slice of my game in the form of a singular level of many. Through this process I have learnt lots about building a game inside a game engine, as someone who specifically specialises in asset creation it was good learning how to actually code, build and test games to a full completion. Furthermore, I learnt processes which also sped up my chosen speciality so I may be more efficient in an actual work setting. I further believe it linked to my chosen RSA brief as it showed how my seagull was not allowed to live in harmony with humans through their decisions and actions.

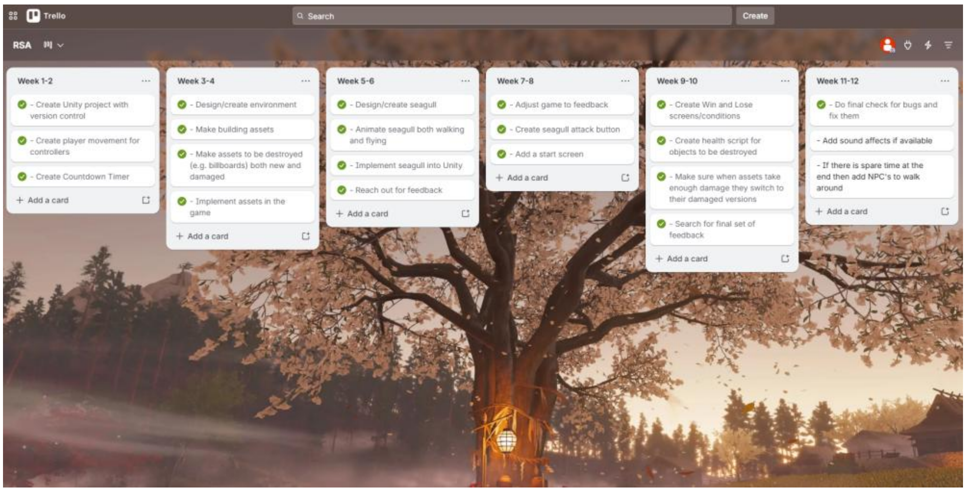

Figure 1

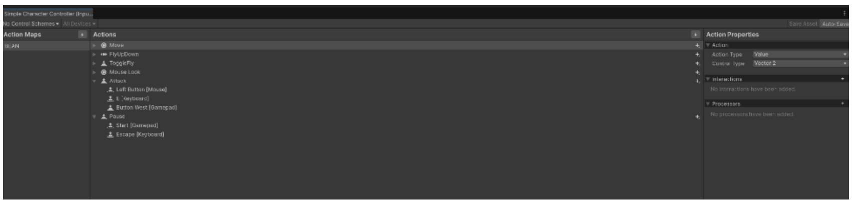

Figure 2

Figure 3

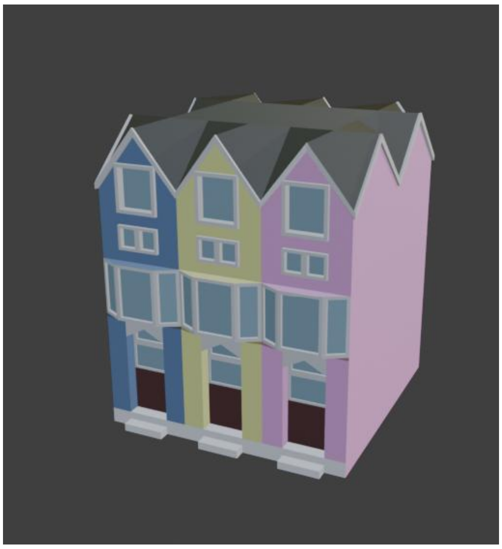

Figure 4

Figure 5

Figure 6

Figure 7 (Wallace & Gromit Wiki, 2026)

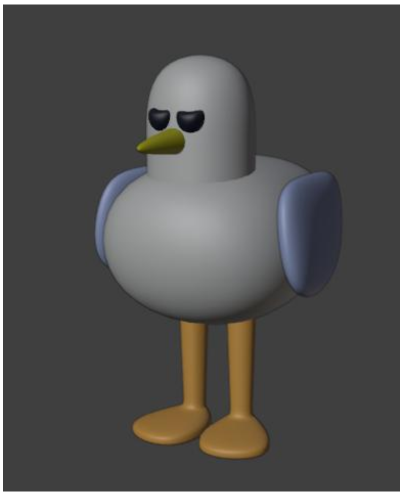

Figure 8

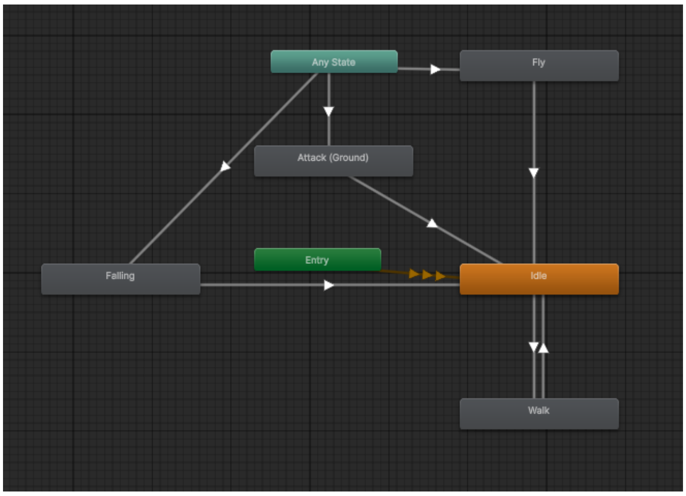

Figure 9

Figure 10

Figure 11

Figure 12

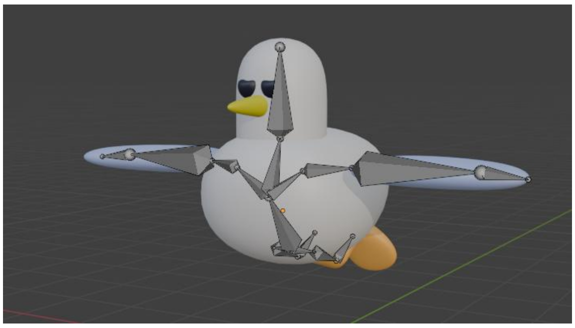

Figure 13

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